Identification of Topics and Opinion Leaders in Reddit Discussions on ChatGPT

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ABSTRACT

Large language models (LLMs) like ChatGPT are spreading quickly and have led to lots of talk online. Reddit is one place where people often share experiences, discuss what these models can do, and learn from each other. This paper looks at how these conversations work. It finds the main topics people talk about and the key users who lead the discussions in Reddit groups focused on ChatGPT. The study uses two methods. It works with a dataset of 52,416 comments. Each comment has its text (comment_body), an ID (comment_id), and a parent ID (comment_parent_id). The study uses Latent Semantic Analysis (LSA) to find and study the main topics in the comment text. At the same time, it uses Social Network Analysis (SNA). This method looks at how comments reply to each other and builds a user network. It uses centrality measures to find the users who have the most influence. The goal is to find out what people talk about when they discuss ChatGPT on Reddit and which users help shape those talks. By combining topic modeling and network analysis, the study shows both what is being said and who is saying it. This helps better understand how online groups form, how influence spreads, and how people share ideas about new AI tools.

Keywords*:* Natural Language Processing, Text Mining, Social Network Analysis, Topic modeling, ChatGPT

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MATHEMATICAL NOTATION

d	text document
D	the corpus of document
G	directed graph
U	an m×m orthogonal matrix, where m is the number of terms
V	an r x n matrix, where n is the number of documents (comments)
d (v, u)	the shortest path distance from node v to node u
A	term-document matrix
N	the total number of nodes in the network
B_i	the collection of nodes that have directed edges pointing toward node i
L_i	the number of outgoing edges from node j
Σ	a diagonal matrix containing sorted singular values

Chapter 1. Introduction

1.1 Problem Statement

ChatGPT may assist in teaching counseling students how to comprehend client circumstances, according to a study by Xie et al. [3]. The results show that ChatGPT is a helpful, interactive, and scalable tool for training counselors. In another study, Ying Lian et al. [1] used text mining to look at 96,435 comments and 55,186 reposts about ChatGPT on Chinese social media. Their research found that people mostly talked about how ChatGPT helps with tasks, its usefulness, and its effects on jobs, education, and technology. They also showed concern about fake information, job loss, and human- machine relationships. Angelina Patience Mulia et al. [2] used a survey to test how easy ChatGPT is to use. They used the System Usability Scale (SUS) and found a score of 67.44. This puts ChatGPT in class D, which means it is slightly above average. Most users said it is helpful, easy to use, and works well, even though it only gives information up to 2021. The study suggests more testing to improve how stable and reliable ChatGPT is. Likewise, Reddit lets users post content, ask questions, and vote on others' posts. Muhammad Y. Hameed BS et al. [4] studied posts from the r/medical school subreddit (from 2009 to 2022). They wanted to understand how medical students think about radiology as a career and what factors shape their views.

Researchers used sentiment analysis with the SiEBRT Robert model to compare 1,542 posts not about radiology with 2,000 posts that were about radiology. The results show that posts about radiology are more positive than posts about other fields ($p < .001$). Positive words included "procedure," "lifestyle," "income," and "tech." Negative words included "AI," "burnout," and "job market." This study helps understand what affects how medical students view radiology and what might influence their career choices. In another study, Amanda M. Bunting et al. [5] looked at Reddit posts from March 5 to May 13, 2020, to see how COVID-19 affected the social lives and behavior of opioid users. They studied 300 posts from r/Opiates Recovery and r/Opiates. The results show that the pandemic changed how

people interacted and led to more online social support. Reddit became an important place for users to share advice about cravings, withdrawal, feeling alone, and recovery. This shows how important online support can be for opioid users during times like the COVID-19 pandemic. Finally, Angelina Patience Mulia et al. [6] studied 713 posts from r/helicopter parents using topic modeling and text mining. Their goal was to understand what people say about helicopter parenting. They found that many posts talked about issues like family, work, college, and education. People also showed concern about social relationships and privacy. The results suggest that parenting should allow children more independence and that more research is needed on how COVID-19 may have affected this parenting style.

ChatGPT has quickly become popular as an AI language tool, and this has led to many discussions on social media, especially on Reddit. People talk about many things, such as how well it works, what it can do, and whether it causes any problems. But because there are so many different conversations, it is hard to find out what the main topics are and who is leading the discussion. This makes it difficult for researchers, government workers, and companies to fully understand how ChatGPT affects society or to make smart choices about how to build, use, or manage it. To solve this problem, need to study Reddit posts. The goal is to find the main things people talk about and figure out which users are shaping these talks. This study is also helpful because it gives a better idea of how people feel about ChatGPT in general. This research can help researchers understand what people expect, worry about, or like about this AI tool by examining the most frequently mentioned topics, such as privacy and data safety so that developers can work to address those issues, this research can show how opinions change over time, which helps people understand how public views on AI grow and shift, which is very helpful for creating messages that match people's feelings and help build trust in AI tools, this identifying the opinion leaders in Reddit talks is important for observing how ideas and information spread in online groups, as these individuals have the ability to influence one another's thoughts and feelings. If companies or government workers talk to these key people, they can share correct information, fix common misunderstandings, and help more people see ChatGPT in a good way. For example, if a popular tech expert says good things about ChatGPT, it can help more people accept and use it, even those who were unsure before. So, working with opinion leaders can make communication efforts stronger and more effective. This research also gives helpful feedback

to developers. It helps them understand what users like, don’t like, and want to change. In this way, the research helps people better understand and live with AI in daily life, this research is also useful for businesses, this helps them improve their marketing plans by showing what people care about most if the study finds that many people like ChatGPT’s creative writing features, companies can highlight that in their ads. Also, by understanding what people are saying about ChatGPT, businesses can find new market needs or chances to offer better products.

1.2 Research Scope

In today’s digital world, online discussions have a strong effect on how people think and form opinions. Places like Reddit have become key spaces where users share ideas, raise concerns, and shape each other's views. The thesis called "Identification of Topics and Opinion Leaders in Reddit Discussions on ChatGPT" matters a lot to many groups like developers, businesses, lawmakers, and researchers. This essay shows why this kind of research is important. It helps understand what people think, make AI tools better, guide rules and laws, and grow what know through studies. This research looks at discussions in the ChatGPT subreddit. A time frame is used to capture many talks so that trends and patterns are seen well over time. Both word and number parts are studied with text analysis tools to find key topics and themes. The words are read to see user experiences, concerns about how ChatGPT works, technical issues, and ideas on what may come next for ChatGPT. The topics are sorted to show what is most talked about and why. The numbers count how often a topic shows up and how much talk it gets. The counts of comments, upvotes, and shares help show which parts of ChatGPT get the most interest and how talks change from the first rollout to later updates.

This study will also look at who the most influential people are in Reddit talks. It will use network analysis to map how users interact to find those who shape others' views. These leaders are picked by how often they post, how much attention their posts get, and how much they change the talk. The study will also use sentiment analysis to see how people feel about ChatGPT. It will look at if posts are positive, negative, or neutral. This will show what users like and dislike and give ideas to improve ChatGPT. The study will also build a network map

to show how users connect. This will explain how people share information and their roles in the talks. The study will watch how people ask and answer questions to learn how users engage. In short, this research will show a full picture of how people talk about ChatGPT on Reddit. It will point out the main topics, key influencers, overall feelings, and user behavior in the community.

1.3 Research Objectives

This study aims to understand how people talk about ChatGPT on Reddit. It will do this by finding the main topics of conversation and the users who influence others the most. The study will look at what people are saying about ChatGPT. It will sort the topics into groups like how ChatGPT works, what users think about it, any problems they face, and what might come next for the tool. Next, the study will check how often these topics come up and which ones get the most attention. It will look at things like how many comments, upvotes, or shares each topic gets. This will show what users care about most when they talk about ChatGPT. Another goal is to find out who the most influential users are. These are the people who post often, get lots of replies, and affect what others think. By using network analysis, the study will find these key users and see how they shape the conversation. This will help explain how opinions are formed and spread in the ChatGPT Reddit community.

Also, the study will check how people feel about ChatGPT. Sentiment analysis will show if talks are mostly good, bad, or neutral. Learning what causes these feelings will help know how happy users are and where to fix things. Another goal is to map the network of talks. A simple map of how Reddit users interact will show how information moves and links in the group. The map will show the ties between those who ask and those who answer. The study will check the roles and behavior of users in these talks. It will split users who ask mostly from those who answer. By studying what they talk about, the study will help understand how users engage with ChatGPT. This analysis will give useful information about what topics interest users, who the main influencers are, and what the overall feeling is in the Reddit discussions about ChatGPT.

1.4 Thesis Organization

In order to identify Reddit's ChatGPT discussion topics and opinion leaders, each chapter focuses on a distinct aspect of the research. The reader is taken from basic concepts to in-depth examination and conclusions in each chapter, the research is introduced in the first chapter, which also gives background information on ChatGPT and explains the significance of online discussions for gathering user trends and views. It gives a short overview of the goals, scope, and research topics. It lists the research questions and ideas and presents the research plan. Chapter 2 reviews previous research on natural language processing, sentiment analysis, and the impact of social media on public opinion; it presents previous research on Reddit conversations and opinion leaders, this providing the foundation for this study. The introduction gives an outline of the research topics and goals. It shows why this study matters for ChatGPT researchers, developers, and marketers. It also shows where past studies fall short. This explains why this research is new and needed. Chapter 3 talks about the research design and methods. It tells why the Reddit subreddit ChatGPT was chosen. It explains how the data was collected. It also shows how to map networks, find opinion leaders, find key topics, and study the text. It lists the tools used in the study and the way sentiment was analyzed. Chapter 4 gives the results. It shows the topics, themes, and main users found in Reddit posts about ChatGPT. It includes numbers like how many topics there are and how many people joined the talks. It also lists the words people used most. It shows how people felt and what they thought about ChatGPT. It includes maps and images to show how users talked and shared information. This chapter also links past research and findings to the questions from Chapter 2. The last chapter gives a short summary of the main results and shows the limits of the study. It gives ideas for more research. It ends with tips on how to make future studies better.

Chapter 2. Literature Review

2.1 Intelligent Text Mining Applications

Yan Liu and the team [9] did important work. They came up with a new way to do smart data mining. Their goal was to make it easier and better to find error data in customer support texts. The method has several parts. One part is the Product Defect Dictionary (PFD). It uses repetition units (GRUs). It also uses association rules (AR). The PFD-GRU-AR model works better than older methods when it comes to mining quality. The way it breaks data into parts works better than doing it all at once. This was shown in a study on service texts about air conditioners. Key numbers that show how well the system works went up a lot. This shows the method is better. The study also found that higher thresholds lead to better results. The best results came when the threshold was set to 0.95. There is also another important topic. The Paris Agreement was made in 2015 at COP 21 to fight climate change. One big issue is transportation. In 2018, this area caused 23% of the world's carbon dioxide emissions. One way to help is by using electric vehicles (EVs). EVs can save money and help the environment. This is why more people are doing research on them, especially in Indonesia. One study focused on electric cars in Indonesia. It looked at how people feel about them, what they like, and what they think. The study also looked at business chances, new technology, and changes in rules. One part of the study was a review of past research. It looked at papers from 2011 to 2023. The goal was to understand what customers are like, what the market is doing, and how EVs affect society. In the end, the study will help create new rules and plans for different groups. The COVID-19 pandemic also caused big changes. One change was the fast growth of working from home (WFH). Many companies made rules to let people work from home. The main reason was to keep people safe during health emergencies. But people still talk about how well WFH really works. Another study looked into this. It used 10,000 tweets about WFH. The study used text mining and sentiment analysis. It wanted to find out what people think is good about working from home. The VADER analysis gave interesting results. The emotion that came up the most was joy [9]. Joy was found in 83.98% of the tweets.

One study looked at an important problem in healthcare. It focused on how the quality of text analysis affects the performance of prediction models. These models help tell if a patient might have health problems later. The study used simulation modeling. The goal was to predict patient deaths in critical care units. It found that even when the text analysis was low quality (F1 score of 0.50), the prediction model still worked well (AUC around 0.80). AUC shows how well the model can tell the difference between patients who will have a problem and those who won't. But there was a problem. The risk predictions were not well-matched with real outcomes. This mismatch is called poor calibration. Calibration means the predicted risks should match the actual risks. Even when a better text model was used (F1 score of 0.80), the calibration was still not great. This shows that while text analysis helps build good models, human help may be needed to make the predictions more accurate [8]. Now looking at the energy sector, researchers have noticed that the hydrogen industry is growing fast. Many countries now treat it as a top national goal. Even though hydrogen energy is becoming more important, not much research has been done on how the technology is improving. One study tried to fix this gap. It looked at patent data using text mining. It used simple tools like phrase search and document vectorization. These tools helped make the analysis deeper and easier to understand. The study showed that the hydrogen industry is growing quickly. Big technical changes were found. One major area of progress was in materials used to store hydrogen. The study also found the development of hydrogen fuel cell cars and better ways to make green hydrogen [11].

This work gave useful tips for better control of BFO crystal growth [12]. Guo Lin and others [16] worked on texture modeling. They used a database method. They used something called generative adversarial networks (GANs). They picked StyleGAN3 to make textures that looked smooth and even. This method made many new textures, even ones not seen in the training data. The team also used a method called GAN inversion to study the textures. They made sure the images matched well with the space that GANs use to create images, called "latent space." They used Versa Stein loss-based steps again and again to help GANs work better with real materials. They also added a way to check how good the textures were. This method looked at how the network acted in small and large parts of the hidden space. They checked the results by comparing them with older ways of looking at texture and structure [16]. In the medical field, Mona Lahad and her team [14] looked at the texture of the pancreas.

Their study looked at people with diabetes. They used CT scans from King Faisal Medical Complex. There were 78 people in the study. This group had 39 people with normal pancreas (control group), 20 with type 2 diabetes (T2D), and 19 with type 1 diabetes (T1D). The researchers took CT scans of everyone and measured 31 structural features. They found that diabetic patients had lower sphericity than healthy people ($p = 0.0246$). Sphericity had 74.36% sensitivity, 69.23% specificity, and an AUC of 0.7538. There was no big difference in sphericity for T2D patients [15]. The researcher studied pills that are hard to dissolve.

2.2 Medical and Health Research

Paula Rowland and others [17] studied how doctors build their professional identities. To study this, they used texts from morbidity and mortality (M&M) ward rounds. Their research showed that M&M rounds are changing. Kim and others [18] looked at the medical tourism industry. They studied 5,507 newspaper articles. These articles were published from 2009 to 2017. They found that daily newspapers and medical journals focused on different things. Daily newspapers talked about medical treatments, support systems, and the tourists themselves. Medical journals talked more about government programs, treatment abroad, and marketing. These different views provide helpful information for making policies. But the study's findings may not apply to all cases because the results depend on the newspapers used. Kun Gao and his team [19] suggested a new method called WEBTRAQIF.

Georg W. Alpers and his colleagues completed research employing automated text analysis on online self-help groups [20]. They wanted to understand what people said in a breast cancer support group. They looked at posts from nine women. The results showed that the LIWC software tool gave similar results to human evaluations. This means the tool works well for this kind of task. However, the tool still needs more work before it can be fully used by researchers or group leaders. Matti Sievert and others [21] studied how well ChatGPT works in the medical field. They tested it on thyroid nodule risk using text descriptions. The results were compared to real lab data and expert opinions. ChatGPT showed some promise. Its sensitivity was 86.7%, specificity was 10.7%, and overall accuracy was 68%. Expert agreement was low at 42%. More studies are needed to make ChatGPT more useful in medicine. Bae and others [22] looked at health literacy in children and teens. They used

network text analysis to study research from the past 30 years. Some main ideas stayed important, such as "health care," "health behaviors," and "prevention." Other ideas like "barriers," "caregivers," and "school environment" have become more important in recent years. The study showed that more research is needed. There is a need for better evaluation tools and school-based programs. Graciela Guerrero and her team [23] looked for an online way to replace the usual anxiety test (AMAS-C). They used text and facial expression analysis to check anxiety in college students. Both methods worked well. Text analysis was a bit more accurate (86.84%) than face analysis (84.21%). This method could help find students who are at risk of failing due to anxiety.

2.3 Economic and Financial Research

Jochen Ruether and his team [24] released a study about Twitter conversations. They focused on how the 2013 taper tantrum event affected asset markets and U.S. monetary policy. They used a method called prospective Dirichlet allocation to study 90,000 tweets. The results showed that tweets contained useful information about asset values. These social media conversations had a strong effect on the term premium of returns. The study shows that social media is playing a bigger role in financial markets. It looks at one past event and a specific data set. Rodrigo Taborda and his team [25] also studied financial communication. They used text analysis to study central bank minutes. They looked at how easy the text was to read, how long the vocabulary was and how often words were used. Their results showed that these communications can shape market expectations. They also help decide the best ways to manage inflation. However, the study only looked at a few countries and used a fixed list of words, which might limit how widely the results can be applied. Still, it shows that clear central bank communication is important. In environmental policy, Kunyan Zhou and others [26] studied how government interest in protecting the environment affected Chinese industrial companies from 2012 to 2021. They used text analysis and they found that when the government cared more about the environment, companies tried harder to lower carbon emissions. Lundong Chen and his team [27] studied China’s decarbonization policy. They looked at about 900 laws from 2006 to 2021. At first, the policies focused on building the ability to lower emissions then they started to control energy use more directly. In the last few years, cutting carbon emissions has become a bigger goal.

Jin and his team [28] studied intellectual property laws to understand the role of policy. They looked at how IPR rules in regional trade agreements (RTAs) affect the quality of Chinese exports. The results showed that adding IPR rules improved export quality by a lot. The exact terms of the agreement mattered more than whether countries followed them closely. The importing country's ability to do research and development played a big part in this improvement. The study clearly shows that intellectual property is important for improving trade quality. But since it focuses only on trade agreements and a few measurements, the findings might not apply in other situations. Bo Huang and others [29] studied how local economies grew in 275 Chinese cities. They focused on the role of party secretaries who had military backgrounds. The study found that these officials reduced government spending, budget deficits, and budget imbalances. It also showed that officials with military experience were more careful and less likely to take risks. This matched the low amount of land given to industry and the low risk seen in issuing junction bonds, which help fund local governments. The study shows why it's important to look at the personal traits of government workers when studying local economies. But it only looked at a specific group of officials and areas in China, which limits how much the results can be applied elsewhere. Suin Lin and his team [30] worked on media and information. They came up with a new way to spot harmful news. Their method used BERT for text analysis and combined it with ensemble learning. The new model was better at finding harmful content. It got an F1 score of 66.3%, which was 7.8% higher than older methods. This made it easier to catch harmful messages. But it is still hard to find complex or sneaky harmful content with high accuracy.

Maya Bogus and her team [31] used quantitative content analysis to study how 159 international organizations are applying the Sustainable Development Goals (SDGs). They found that although many organizations mention the SDGs, real policy integration is still missing. This means that many groups continue to work independently rather than in a unified way. More research is needed to understand the barriers that prevent better policy coordination. YuGu Zhang and others [32] studied how the construction industry in China dealt with the COVID-19 pandemic. They talked to professionals from Chinese construction companies and looked at 1,572 annual reports. They found 19 main themes of resilience and made an Operational Asset Framework (OAF). Gong and his team [33] studied something different. They looked at TripAdvisor reviews to sort Cultural Ecosystem Services (CES) in 48 U.S.

national parks. They put the services into four groups: beauty, spiritual or nature value, identity or fun, and cultural, learning, or social value. These results can help with park planning for protection and tourism. But using only TripAdvisor reviews is a limit. The data may not show what all visitors think. Wen Sheng Yu and others [34] studied 84 noise control policies in China from 1989 to 2022. They found the use of policy tools was not balanced. They said the country should improve the mix of tools, make policies steady, and add more flexible ways to act. Most policies used only strict rules from the government. There was little use of money-based tools. The study also said not enough was done to share or enforce the rules. Noise from traffic and factories was not handled well. For example, Xiaolan Chen and others [35] looked at energy policies in 29 Chinese provinces from 2006 to 2020. They wanted to see how China deals with energy poverty.

2.4 Technological and Computational Research

Mika Banhala and others [36] studied how research changed from 2000 to 2018 on using large datasets to understand online consumer behavior. They used modeling and text analysis to find important papers, well-known authors, and how research topics shifted over time. Wenxiong Liao and others [37] took a different path. They made a new way to study sentiment using a multilayer graph neural network (MLGNN). Usual GNNs only look at nearby text features. MLGNN looks at both close and wide-ranging data. It does this by using a better dot-product attention tool and different ways of connecting nodes. Tests show that MLGNN works better than older models in sentiment analysis. Fei Lin and others [38] worked on emotion analysis. They created Style BERT, a new model for studying emotion in both text and speech. Style BERT tries to make the meaning of text and speech clearer before putting them together. It learns the different "styles" of each type. One special part of the model is the Interactive Style Enhancement Module. It uses logmel spectrograms to find small details in sound. Style BERT does better than older models on standard tests. But it may be hard to use because it needs advanced tools and complex setups. Yuting Lin and others [39] made the Lifetime Text-to-Audio Affect Analysis (LTASA) model. This model learns to understand emotion in speech and text over time. It finds meaning between and inside both types of data. It uses a special tool called a complementarity subspace and a knowledge dictionary made for each type. Tests on three datasets show that LTASA does better than basic models.

Yuming Lee and others [40] gave a framework that ends in an emotional analysis. This framework is based on psychological ideas. Its main goal is to find where emotions in the text come from and what causes them. The goal is to help people trust deep learning results more. The method does this by pulling out and showing phrases that express strong feelings. This makes the results of sentiment analysis stronger and easier to understand. Livia Cerrado and others [41] suggested a new way to group text. They used a tool from social network analysis that many people use. The method looks at which words appear and how they are spread out. It shows text as a network of linked items. Tests show that the method works well. It also works better than some older methods that use vector models. These network-based methods give new ideas and may make grouping more accurate. Zhang Hejin and the research team [42] made a type of AI that people can understand. This model checks how original a patent is. The model's biggest strength is that it includes the structure of patent claims. It also helps explain how to find new ideas. The team trained the model with a dataset from the European Patent Office. They tested it on patents about in-vehicle communication networks. It worked better than older models. H.T. Ha and others [43] built the OCRMiner system. This system helps pull out information from scanned invoice images. The system makes use of layout and text analysis tools. The objective is to use these semi-formatted documents to automatically create metadata. In essence the technology imitates how people read. Text characteristics and layout data are changed with a 90% data retrieval rate for English invoices and an 88% rate for Czech invoices, the tests demonstrated significant efficiency.

To find important ideas and research trends in communication policy journals, Kwon and others [44] looked at the abstracts from these journals over 40 years. The data show that differences in research results between continents got smaller over time. But there are still big differences between countries. The data also show that the focus of journal research changed a lot around the mid-1990s. This study mostly looked at keyword frequency, which might be a weakness. Still, it gives useful information about how the journal's research has changed. A similar study was done by Lev Sarkisov and others [45]. They used text mining to study how nanomaterials are made. They wanted to find a way to collect and use shared knowledge from scientific papers. Their case study looked at 1600 synthesis processes, with a focus on ZIF-8. This helped them find patterns and links between different synthesis conditions. It also made it easier to compare methods without doing it all by hand. Finally, V. Vasileev and his team

[46] studied open databases. These databases have information on software problems, security issues, and cyberattacks. Their study showed that using text mining on these texts can help. They suggested an automated method. This method uses semantic analysis to study software risks in industrial control systems (ICS). They showed how it works with a case study of an oil plant. This method can find certain risks.

Chapter 3. Methodology

3.1 Term Frequency-Inverse Document Frequency

To figure out the main topics people discuss in Reddit threads about ChatGPT, this thesis uses the Term frequency- inverse Document Frequency (TF-IDF) method. TF-IDF is a standard technique in text analysis. It helps measure how important a specific word is within one document, especially when comparing it to a larger collection of documents, which this called the corpus. In this research, treat each Reddit post or comment as one "document." All the ChatGPT-related posts together make up the "corpus." TF-IDF has two parts. The first part is Term Frequency (TF). TF counts how often a word (call it "t") appears in one document (call it "d"). Then this number is adjusted, usually by dividing it by the total number of words in that document. This step is called normalization. It is important because documents have different lengths. Normalization helps make the TF scores fair across all documents. TF gives more weight to words that appear often in one document. This means the word is likely important to that document’s topic. For example, if the word "ethics" shows up many times in a comment, a high TF score means it is likely important to that discussion. Now comes Inverse Document Frequency (IDF). IDF checks how often or how rarely the word "t" appears in the whole corpus "D." It uses a formula. The IDF is the logarithm of the total number of documents (D) divided by the number of documents that contain the word "t." By using logarithms, the results are smoothed down and the influence of very common or uncommon words is minimized. IDF may be used to emphasize words that are distinctive and could imply a specific subject, as well as to effectively boost the prominence of somewhat rare phrases in all Reddit posts. Words that are often used such "the" and "a," get lower IDF ratings, on the other hand, words like "bias" and "performance" are less common overall because the lesser the frequency of recurrence, the higher the IDF score, these terms are useful for discovering original debate topics. The TF value of a phrase in a document is multiplied by its IDF value to get its TF-IDF score. The most important debate themes are indicated by this total score, which assists in identifying phrases that are significant in a certain post yet comparatively uncommon in the conversation as a whole.

$$\text{tfidf}(t, d, D) = \text{tf}(t, d) \times \text{idf}(t, D) \quad (1)$$

From (1), to understand the main topic of a Reddit post, the researcher first needs a way to identify the most important terms in the discussion. The researcher accomplishes this by calculating the TF-IDF score of the words used. This score helps identify the terms that best represent the content; the TF-IDF value itself is calculated by multiplying two numbers: the term frequency (TF) plus the inverse document frequency (IDF). In this work, the researcher treats each individual Reddit comment as a separate document. All comments constitute a corpus, or entire collection of textual analysis. The researcher computes a TF-IDF value for each word in this corpus. The unstructured text comments are transformed into a structured numerical matrix using this procedure. Each column in this matrix corresponds to a distinct word in the complete lexicon, and each row represents a remark. It successfully eliminates often used stop words, such as "the," "and," "is," and so on.

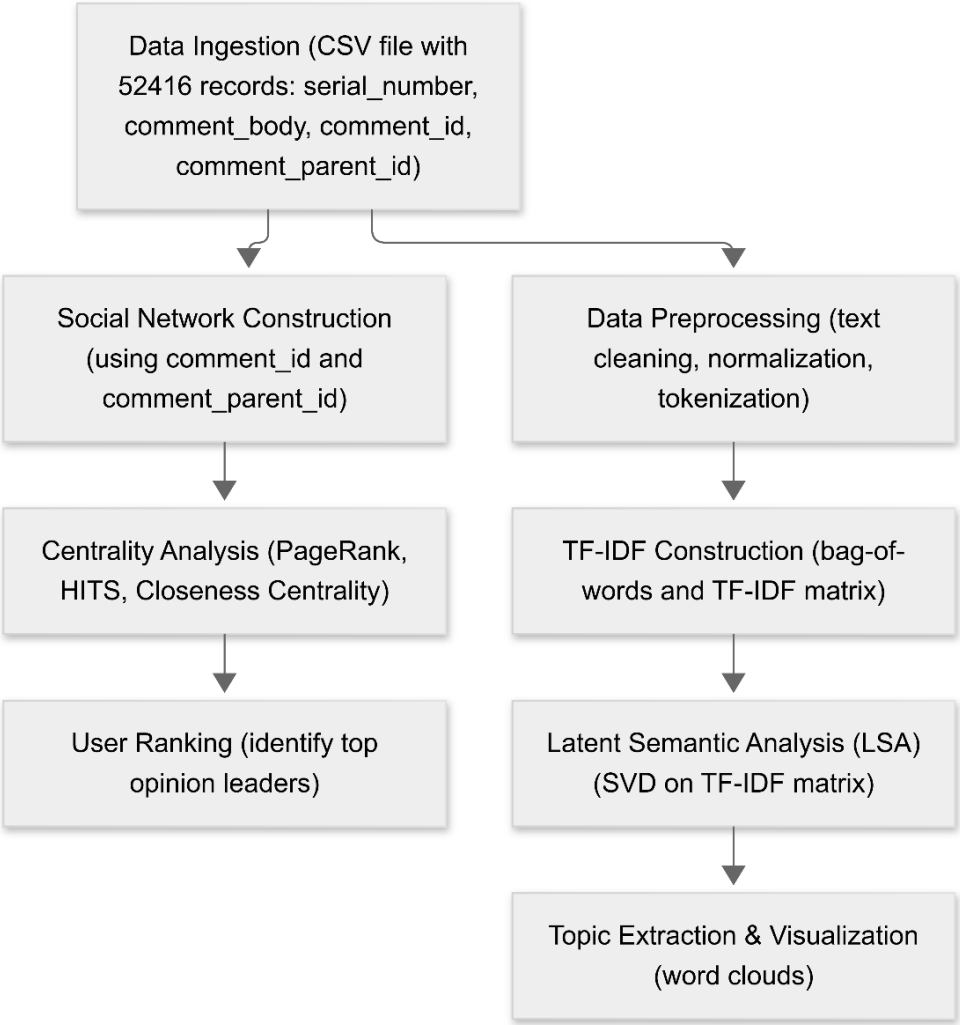

Figure 3.1 System Architecture

3.2 Social Network Analysis

Finding key users, also called opinion leaders, is an important step, as shown in Figure 3.1. Another goal is to understand how people talk to each other in these GPT Reddit discussions. This research uses social network analysis (SNA). SNA provides a way to measure and study how people connect and interact. In this method, each person is shown as a node, or point, in a network graph. The lines, or edges, between these points show replies and other user interactions. Once the network is built, the researcher can use several centrality tools and methods. These tools help find users who guide discussions, share a lot of information, or are seen as experts. The Reddit dataset includes comment_id and comment_parent_id. These give the needed data to build the network. They also make it possible to track the structure of replies

in each conversation. Making a social network graph with this data is not hard. All you have to do is find each Reddit user who left a remark in the dataset.

Every user becomes a network node, also called a vertex. These user nodes are connected by directional edges, or links. The response system in Reddit's comments is the only way these edges are made. A directed edge is created if one comment (identified by comment_id) is a reply to another comment (identified by comment_parent_id). This edge connects the person who replied to the person who made the original comment. This directed link shows a direct connection and it means information might flow from the person replying to the original commenter. This user-level interaction network based on replies from a ChatGPT conversation is shown in the graph. It should be noted that each comment_id and the author associated with the comment_parent_id are crucial to creating this user-centric network structure, even if the comment_body is mostly used for modeling themes (see Section 3.4).

$$G = (V, E) \tag{3}$$

This study uses the comment_id plus comment_parent_id information to map out the reply relationships. This mapping forms the directed graph, which is called G. In this graph G, every unique user is represented as a node, each reply creates a directed edge pointing from the user who wrote the reply to the user who wrote the original comment being replied to and then calculate several key SNA metrics assess a user's influence plus position within this network. They focus on different aspects of centrality alongside structural importance. Degree Centrality is a basic measure. It counts how many direct connections a user (node) has. Since the network is directed, the researcher needs to make an important distinction. In-degree counts the number of replies a user gets. Out-degree counts the number of replies a user gives. A high in-degree is important. It shows that the user often gets replies. This is a sign that the user may be an opinion leader. It means the user’s comments often get responses or attention from others. Users who get many replies (high in-degree) often become central figures in the discussion. Out-degree shows how active a user is in replying to others.

$$Cc(v) = \frac{N-1}{\sum_{u \neq v} d(v,u)} \quad (4)$$

In addition to direct connections, Closeness Centrality which is the inverse of the average shortest route distance between a user and every other accessible user in the network, measures how simple it is for a person to get in touch with every other user because their effective network positioning, users with high proximity centrality may be able to swiftly spread their knowledge or viewpoints across the discussion community. The shortest route distance (number of edges in the shortest directed path) between nodes v and u is d (v, u), where N is the total number of nodes in the network (or linked component), on average, they are 'close' to everyone else, indicating they play a major part in the flow of communication rather than being restricted to side discussions. Users are well-integrated and have the potential to be successful broadcasters inside the network, are identified by this statistic.

$$a_i = \sum_{j:(j,i) \in E} h_j \quad (5)$$

$$h_i = \sum_{j:(i,j) \in E} a_j \quad (6)$$

Explicitly identifying between people functioning as major pointers to information and those who are the originators of that useful material, this research applies the Hyperlink-Induced Topic Search (HITS) algorithm. initially intended for the web, HITS adapts nicely to interaction networks like Reddit responses. Each user node is given two unique scores: a Hub score and an Authority score. Authorities are people whose remarks are answered by several knowledgeable and engaged Hubs, a person with a high Authority score is one whose contributions are valued by people who often interact with influential material, Hubs are users who respond to many Authorities with high scores. A person who actively engages with these reliable sources and helps to focus attention throughout the network is indicated by a high Hub score and an iterative procedure is used to generate these Authority plus Hub scores, which reinforce one another until the numbers stabilize, a node's Authority score affects the Hub scores of those connected to it, and vice versa. These discrete responsibilities within the conversation network are more precisely identified thanks to this recurrent refining.

3.3 Page Rank

This study also uses the PageRank algorithm to identify influential comments and, by extension, influential users within the ChatGPT Reddit discussions. PageRank was originally created by Larry Page plus Sergey Brin. They developed it to rank web pages based on how interconnected they are and adapted this algorithm for the Reddit discussion network. In setup, each unique remark becomes a node (a point) in the network. A directed edge (a connection) is formed from a reply remark to the parent comment it responds to this framework enables to locate the comments that have substantial impact inside the broader discourse, the underlying concept behind this is simple, when someone reacts to a remark, take it as a sort of endorsement or continuance. This reply distributes a percentage of "influence" from the responding remark back to the original parent comment. The entire network is modeled as a directed graph, G = (V, E). Here, V represents the set of all comment nodes, plus E represents the set of directed reply edges connecting them. With this network built, the PageRank score for any comment node tells something specific. It reflects the probability that someone randomly following the reply links would eventually land on that comment. Put simply, comments with higher PageRank scores are considered more central or influential within the discussion network.

$$P_i = \alpha \sum_{j \epsilon \beta_i} \frac{P_i}{L_j} + \frac{1-\alpha}{N} \quad (7)$$

The researcher calculated the PageRank scores. This uses a repeated process. The algorithm repeats steps, improving the scores each time. It keeps going until the scores stop changing much from one step to the next (this is called convergence). A key part of this process is the 'damping factor' (written as α). This factor helps make sure the scores stabilize. It also gives each comment a small base level of importance, even if it doesn't get many replies. It understands that important comments can come from those not directly linked but still have a big effect on the discussion. The PageRank algorithm looks at comments that reply to a specific comment (let's call it comment i). The group of comments that respond to comment i is important (often shown as B_i). The idea is simple: if many comments reply to comment i, it means comment i has influence in the conversation. These replies help support or link back to comment i. This group of replies is important. It shows how the algorithm gathers the 'influence' passed down from earlier comments to calculate the PageRank score for comment i. Another important part of the PageRank calculation is the number of replies a comment makes. This is shown as L_j, where 'j' is the comment that is replying. This value shows how much influence comment 'j' spreads to the comments it replies to. If comment 'j' replies to many comments, the influence it gives to any one comment becomes smaller. This dilution is important. It stops comments that reply a lot from unfairly increasing the scores of all the comments they reply to. Instead, the influence of each comment is divided by the total number of replies it makes its. The total number of comments in the network, which the researcher call N, is also an important factor.

This value mainly works together with the damping factor (α) in the PageRank equation, it helps ensuring every remark receives a basic, minimum PageRank score, even if no other comment immediately responds to it. This represents the slight risk that someone viewing the forum would randomly jump to any remark. This technique is significant since it prohibits comments from obtaining a zero score merely because they are isolated or less linked. It also adds to keeping the total PageRank computation steady plus ensuring it converges appropriately, the damping factor, α when this number is commonly set at 0.85, result shows the likelihood

that a hypothetical user, randomly surfing the debate, will follow a reply link rather than merely moving to a totally random remark elsewhere. In order to support this theory, the researcher explicitly applies these ideas to the ChatGPT debates on Reddit. The researcher calculated PageRank scores for each comment by modeling the reply relationships as directed edges between comments and their authors. A high score suggests they have a major influence on the general direction and flow of the debate because their comments are regularly mentioned or responded to.

3.4 Latent Semantic Analysis

Following work with Term Frequency-Inverse Document Frequency (TF-IDF) scores, which is detailed back in Section 3.1, the next logical step is using Latent Semantic Analysis (LSA). The researcher applies LSA here specifically to uncover the underlying, or 'latent', topics present within the Reddit discussions about ChatGPT. LSA is a statistical method that uses a strong math tool called Singular Value Decomposition (SVD). LSA takes the big term-document matrix made from the TF-IDF scores and makes it simpler (reduces its size). This is important because it helps show hidden connections between words, how words relate to each other, and how posts are connected based on the words they have in common. For this thesis, LSA helps group Reddit posts into clear topics by finding patterns of words that often appear together in different posts. This improves the initial results from TF-IDF and gives a clearer picture of the main topics. LSA starts by using the term-document matrix (which the researcher can call matrix A) made from the TF-IDF scores. In this matrix, rows are terms and columns are the Reddit posts. The core goal of LSA is then to decompose this matrix A. It aims to break it down into two more meaningful matrices: one linking documents (posts) to the identified topics and another linking those topics back to the specific terms (words) that define them.

$$A = U\Sigma V^T \tag{8}$$

As shown in formula (8), the researcher uses SVD to break the term-document matrix A into three parts. Matrix Σ is a diagonal matrix with sorted values. The biggest values show the main

topics in the dataset, like technical abilities or ethical issues. Smaller values usually show less important topics or noise. These parts work together to find hidden topics and their connections to the words and comments. Matrix U links each Reddit comment to hidden topics. Each column shows how much a comment is related to a topic. A comment with a high value in a column is strongly connected to discussions about ChatGPT's performance. Matrix V^T connects words to these hidden topics. Its columns show how important specific words are to a topic. A high value linking the word "accuracy" to a topic means that the word is important for describing that topic.

In LSA, truncation is an important step, when the top 'k' biggest singular values (from Σ) and the matching portions of the U and V^T matrices are retained, then the original data is reduced in dimension and simplified. The way eliminate unnecessary information and noise is the primary advantage, this condensed representation highlights the most essential patterns of how words tend to appear together across the many comments, making subsequent research much easier, the degree to which comment 'd' is related to subject 'i' can be determined by examining the values in the shortened U matrix (let's say U_{di} for document 'd' and topic 'i'), the importance of word 't' in determining topic 'i' may be shown by looking at the values in the truncated V^T matrix (V_{ti} for term 't' and topic 'i'). Each dimension in this space relates to a hidden topic that is characterized by a distinct combination of powerful phrases, the researcher separate the main conversation topics from the intricacy of the initial high-dimensional TF-IDF matrix, this transformation is crucial for subsequent stages, the role of opinion leaders may be analyzed since it is possible to trace the highlighted themes back to comments and, consequently, their writers, people are likely to be considered significant figures in the ChatGPT Reddit community if their comments are highly related to a number of popular themes.

Chapter 4. Experiment

The experiments used a high-performance computer with an AMD Ryzen 7 8845H processor and 32.0 GB of RAM. All programs were executed and analyzed on MATLAB version R2024a, which served as the main platform.

4.1 Text Data Processing

The data for this study was obtained from Kaggle [47], where the raw text data consisted of 52,416 Reddit comments related to ChatGPT discussions stored in the Comments_body field. A methodical approach to text processing was necessary to get this data ready for topic modeling and further analysis. The researcher cleaned up the raw data and using term frequency-inverse document frequency (TF-IDF) weighting are the two primary procedures in text data processing that are covered in this section. MATLAB is selected as the primary tool to complete the assignment because of its strong text analysis features and capacity to handle big data sets. A tokenized document function is used to partition captured text, converting each annotation into a collection of lexical tags.

The researcher used previously tokenized comment content, and this function helped build a bag of words model. A term frequency-inverse document frequency (TF-IDF) vector was calculated for this study using the MATLAB function bagOfWords. The weight of each word in the lexicon is shown in the resulting matrix. This TF-IDF calculation was done specifically on the review data's text. It checks how important each word is by looking at how often it appears across the data and how often it appears in a specific text. The researcher set the parameter TF-Weight, and the term frequency was scaled using a logarithm. The formula used was TF = log (1 + original word frequency). The effect of very common terms is lessened because to this logarithmic scale. This helps manage any word repeats and lessens the overall effect, in lengthy publications, the inverse document frequency is determined by the IDF-Weight option and the conventional method is used. Rarer maybe educational words are employed to counterbalance the impact of very common terms, this is equivalent to the traditional implementation of TF-IDF.

As shown in Figure 4.1, the TF-IDF matrix produced by this procedure has 51864 documents and 46663 unique words.

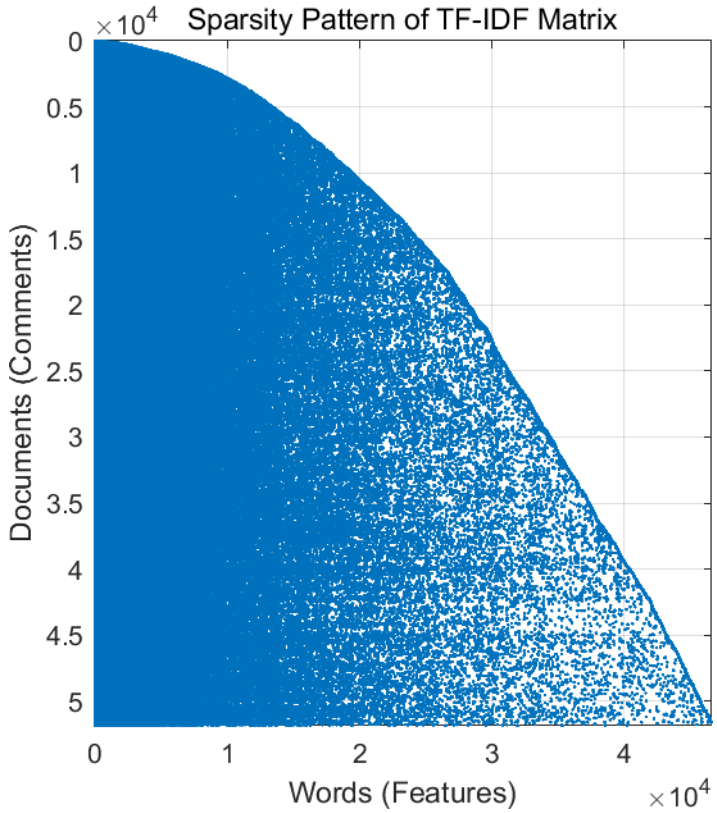

Figure 4.1 Sparsity Pattern of TF-IDF Matrix

A raw data collection was made for the early stages of this experiment. This step was needed for the next network analysis, and the data came from the "chatgpt-reddit-comments.csv" file. This preprocessing was important. It helped make the data consistent and accurate, which allowed for the correct creation of the comment interaction graph. The dataset was loaded into the MATLAB environment. It included 52,416 entries with IDs, parent IDs, and comment text. The MATLAB function Read Table was used here. This method helped parse the CSV file and organize the data in a table. The researcher added a new serial_number column to the database as the first step to organize the data. A unique, sequential integer between 1 and the total number of comments was given to each comment. This provided an easy index to refer to specific entries. Cleaning the comment_parent_id column was an important part of the data preparation. The initial checks showed that prefixes were in the parent identifier. Prefixes like "t1_" or "t3_" are made up of two alphanumeric characters followed by an underscore and are part of Reddit's internal system. These prefixes could cause problems for mapping comments and parents in the

network graph, even though they are useful on Reddit. In the graph, each node should only have the ID of the central post or comment. So, a cleaning process was set up. The code used a regular expression function to check each element in the comment_parent_id column. The target pattern was two alphanumeric characters ([a-zA-Z0-9]{2}) followed by an underscore (_) at the start of the ID string (^). The researcher removed the prefix, leaving only the part of the ID needed to build the network. This cleaned-up file was used as the input for creating the network graph and performing the centrality analysis described in detail later in this chapter.

4.2 Implementation and Evaluation

Once the Reddit data has been processed and these centrality metrics have been computed by the researcher, rank and score people according to their influence ratings. The N most influential users (set N to 5 for this experiment) are chosen after the page rank values are sorted in decreasing order. The rating gives clear information that helps find the most influential users. It is useful for more research, topic modeling, and sentiment analysis to understand the link between network impact and conversation content. Visualization is part of the assessment, but the main task is still computing and sorting the centrality values. The code also includes visualization features like force-based graphs and bar charts. Key opinion leaders are shown on the graphs, which help make the findings easy to see and understand. These visual aids make it easier to understand the network's structure and the effect of different users. The Figure 4.2 displays a force-directed network graph where each node represents an individual comment or parent reference from the Reddit dataset, and the edges indicate reply relationships between them. The visualization organizes nodes into loosely defined columns and clusters, reflecting the inherent structure of threaded discussions. Dense regions within the graph indicate active conversation chains with multiple replies branching from a common comment, while sparser areas suggest isolated or less interactive threads. Overall, the layout provides a visual overview of the hierarchical and branching dynamics of the comment threads, revealing both the diversity and interconnectedness of the discussions.

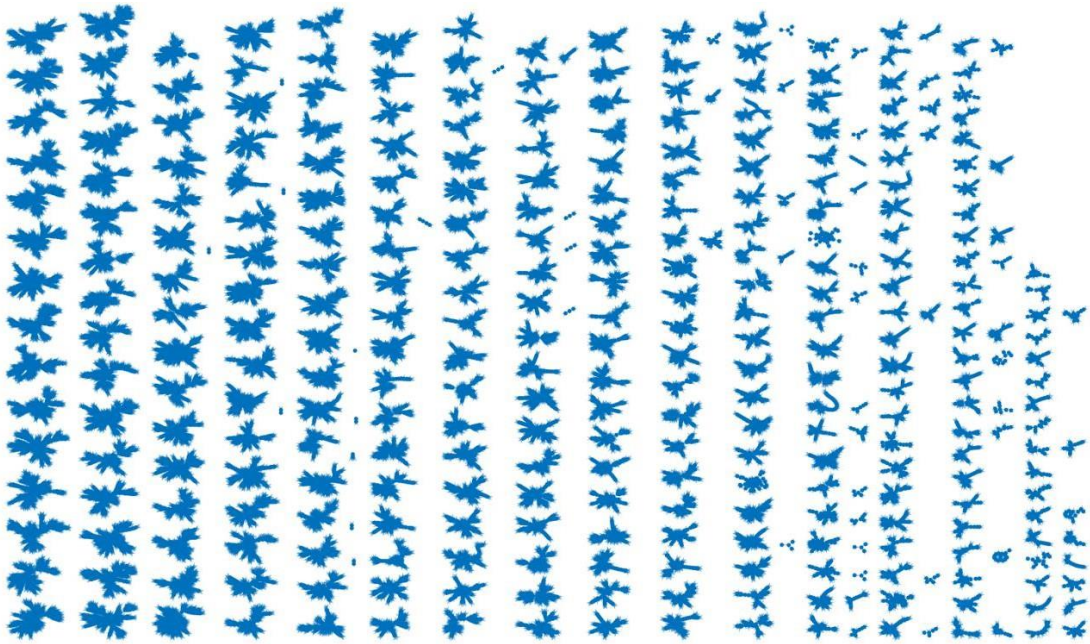

Figure 4.2 Comment Thread Graph

The page rank analysis's conclusions are clearly shown in the bar chart in Figure 4.3, which shows that five comments had the highest page rank values. The comment identified as 107mr00 earned the highest score, approximately 0.0030. Following closely are 107vs7i (around 0.00285), 10heg9p (around 0.00275), 10gzw93 (about 0.0026), and 10mias6 (near 0.0024). In terms of the network response structure, these five comments are the most important discussion centers or are identified as the most influential posts in PageRank. The hub-centricity score was also calculated and graphed. The results show the top five users based on this score, including j1vrpdg, j1vi6k, and j1vuu7f. These users are the most important nodes in the network. This means they interact with many other commenters, especially those seen as authorities. The analysis shows that the scores for these top hubs are very close. This suggests that these users mainly act as key connectors or information intermediaries in the network and play an important role in the structure of the discussion.

The determined centrality ratings served as the foundation for identifying important viewpoints that contribute to a particular structural function in the conversation as well as possible opinion leaders. The nodes were sorted in descending order based on the score for each metric. The analysis focused on identifying the top five comments for each centrality type. The results of the node centrality analysis are shown in the bar chart in Figure 4.4. The five comments that achieved the highest hub values are highlighted in this figure. They are comments J1VFPD9,

J1VIL6K, J1VIUWF, J1VMJ72, and J1W8DLQ. It is worth noting that the stroke scores are all very close to the specified maximum (approximately 1.5 x 10^{-4}), in Figure 4.4, this bar chart shows the top five comments in order according to their centrality score. The vertical axis represents the authority score, and the horizontal axis shows the unique comment id for each comment, which serves as an identifier: the comment with ID 107mr00 has the highest authority score (about 0.0147); 107vs7i, 10gzw93, 10heg9p, and 10gy5dx (about 0. 0115) follow. The relatively high scores indicate that these comments are structurally important when compared to most other web comments that are closer to 0. These comments represent the point at which the discussion converges. These posts elicit the most direct responses and are positioned as the central content in each thread of the discussion. The show comments are classified as interesting or important by other participants and these contents and their creators are likely to be investigated as influential elements or opinion leaders within the ChatGPT Reddit discourse.

In figure 4.6, Comment 107mr00 exhibits the highest In-Closeness score (approximately 0.9 x 10^{-7}), followed by 107vs7i, 10heg9p, 10mias6, and ztd950, all with scores clustered between roughly 0.74 x 10^{-7} and 0.81 x 10^{-7}. A high In-Closeness score suggests these comments are structurally central within their respective discussion threads, meaning they can be reached relatively quickly from many other comments when moving up the reply chain. Such comments summarize the most important points of the discussion and may also be important arguments mentioned by many others. The “out closeness” value in Figure 4.7 shows that it is much lower than the “in closeness” value, about 7.0 x 10^{-10}. The top five comments are j68hshb, j66ciin, j66cewa, j658cmd, and j653u62. The values of these top comments are very similar. The high out closeness means that, in the graph's structure, many other nodes can be reached easily by moving down the answer tree (in the opposite direction as the edge). This shows that these comments are likely used as a starting point for conversation or as early answers on a broader topic. These comments are also high in the comment hierarchy. This position helps them "broadcast" or sift through many follow-up comments in their own discussion area. The large difference between intimacy and non-intimacy scores likely shows how Reddit threads branch out. It may take going through many comment branches to find a key comment (high intimacy), while the original comment may flow down into larger, more direct branches to reach many other comments (high parent density).

The researcher added an interactive analysis feature to look more closely at the content of specific influencer comment threads. The graph has a callback function called onNodeClick. When a user clicks on a comment node, this function does several steps. It finds the comment ID of the clicked node then finds all comments that depend on it. Finally, it determines the scope of the discussion starting with or following that comment. To do this, temporarily flip the direction of the graph's edges to flip edge (G), so that the edges face from parent to child. Then, it performs a breadth-first search (bfsearch) starting from the clicked comment's ID to find all reachable descendants. Third, the function filters the original dataset. It retrieves the actual comment_body text for the initially clicked comment plus all the descendants it just identified. This subset of text represents the specific sub-conversation that will be analyzed for its topic. The core of the topic analysis, happening inside the onNodeClick function, involves text preprocessing followed by Latent Semantic Analysis (LSA). The collected comment texts from the sub-conversation are processed first. This includes changing all text to lowercase, removing punctuation, and deleting common English stop words. MATLAB functions like tokenizedDocument, erasePunctuation, and removeStopWords do these tasks after the cleaned documents are used to create a bag of words model. The TF-IDF weights in this model help show which words are rare in the overall subdialog but appear often in this one. The TF-IDF matrix is calculated using tfidf(bag). This draws attention to terms that are relevant to this subject and sparse TF-IDF matrix is then subjected to a singular value decomposition (SVD) using the svds(tfidfMat) function and LSA, SVD breaks down the term-document matrix into its underlying semantic dimensions or themes. The first right-most singular vector (V(:,1)) is the first possible subject that is extracted in this method. Lastly, the sub dialog vocabulary's words are arranged according to the size of their weights in this subject vector. By clicking on the user-selected primary subject of the discussion thread, this step finds the words that are most pertinent to it.

This research looked at the content of sub-dialogs linked to the most highly rated comments using PageRank analysis. Figure 4.8 shows a word cloud for the discussion related to comment 107mr00. This comment was found to be a very central node. The word cloud shows a clear topic. The words "symptoms," "OCD," "ADHD," and "randomization" are important. Other weighted words like "control," "trial," "study," "treatment," "nac," "glutamate," and "amino" also help show the topic. These words together suggest the topic is about clinical research or

treatments. The terms point to a discussion about randomized controlled trials. These trials may be testing treatments using things like N-acetylcysteine ("nac") or other amino acids that affect the glutamate system. The context appears to be the management of symptoms related to disorders like attention-deficit/hyperactivity disorder and obsessive-compulsive disorder. Smaller words like "potentially," "improve," and "children" give context and imply a discussion of the study's findings and the target patient group. Another word cloud is shown in Figure 4.9. The words "must" and "change" stand out. These two words appear often in the comments for this topic. Their size in the word cloud shows they are important. These words are likely about ChatGPT. They point to themes like need, action, and change. Other medium-sized words include "change," "sit," "together," "make," "break," and "chain." These words add more detail to the main idea. They may describe how change happens ("break," "make"), the need for group effort ("together," "unite," "speak"), or pushing past barriers ("chains"). It's intriguing how words like "chorus," "full chorus," and "poetry" have emerged and these terms can sometimes be used literally to describe a song's structure, but other times they serve a metaphorical purpose in technical discussions when talking about creative songwriting or music generation with ChatGPT.

The word "code" is very noticeable when examining the word cloud in Figure 4.10. This word's prevalence suggests that it plays a significant role in the debate of this subject and the words "chatgpt" and "probably" are related in both size and meaning implies right away that the main emphasis is on ChatGPT's functionality, constraints, or significance in relation to software code. The word "probably" shows that these talks may involve guesses or indirect ideas. The word "programmer" is also important. It shows that the role or opinion of the human programmer matters in these talks. Together, these key words suggest that the comments are about how software development and AI language models connect. Figure 4.11 shows a word cloud where most words are about time or amounts. Words like "per case," "month," and "average" stand out. These words appear often and seem to matter a lot in what the user said. The word "maintain" also appears, which points to keeping things steady or managing lasting costs. This group of words points to talks about fees, regular costs, math, or long-term money planning. Medium words like "person," "month," "house," "car," "salary," and "minimum" support this idea. These words show that the focus is on budgeting, living costs, and managing money. The talks seem to deal with income and costs on a month-by-month and person-by-person level. They mention

basic needs like a house and a car and include pay levels like "wage" or "minimum wage." Some words are very exact, like "Toyota," "Corolla," and "flatmate," along with numbers like "1000" and "725." The word "scenario" itself makes it clear that the discussion and modeling of fictitious financial conditions are the main emphasis. In the case of user 10gzw93, the words "ChatGPT," "AI," or other technical parts of the model either appear clearly or are missing in the word cloud. This is very different from the last case. It shows that even though this user is active in the wider ChatGPT network, the main points in their comments are mostly about money. The talks seem to focus on cost-benefit thinking and personal money management. These money-related talks may happen in threads about how AI affects jobs and income. The actual expenses of having access to technology and related lifestyles might be the topic of debate instead. The tool itself may not be the most common phrase in these specific comments, it is probable that users are talking about using ChatGPT for financial planning and analysis (see Figure 4.12). The postings of user 10mias6 that the fifth-most significant participant based on PageRank are represented by this word cloud. Key opinion leaders in the ChatGPT subreddit are leading a distinct, more specialized subject of debate, as seen in this image. The words "submission," "job," and "job" are the most common ones in the word cloud. This shows a strong focus on job-related tasks. These include writing and sending applications or talking about job openings. Other common words like "interest," "skills," "show," and "cover" also point to this main idea. The word "cover" likely means a cover letter. These words suggest people are trying to show they are qualified for jobs. They also point to finishing the whole application process. This view is expressly supported by words like "write" and "apply." These terms suggest a conversation about active job searching and application writing. Words like "person" and "project" imply that the topic of discussion may be speaking with a recruiter giving a presentation regarding prior work, or working together on a professional project. Interestingly, smaller terms such as “basically,” “nice,” and the informal term “crap” suggest a potentially pragmatic and potentially critical or conversational tone regarding job search experience, application requirements, or the effectiveness of a tool like ChatGPT in a field suggest that the following is a good of a tool.

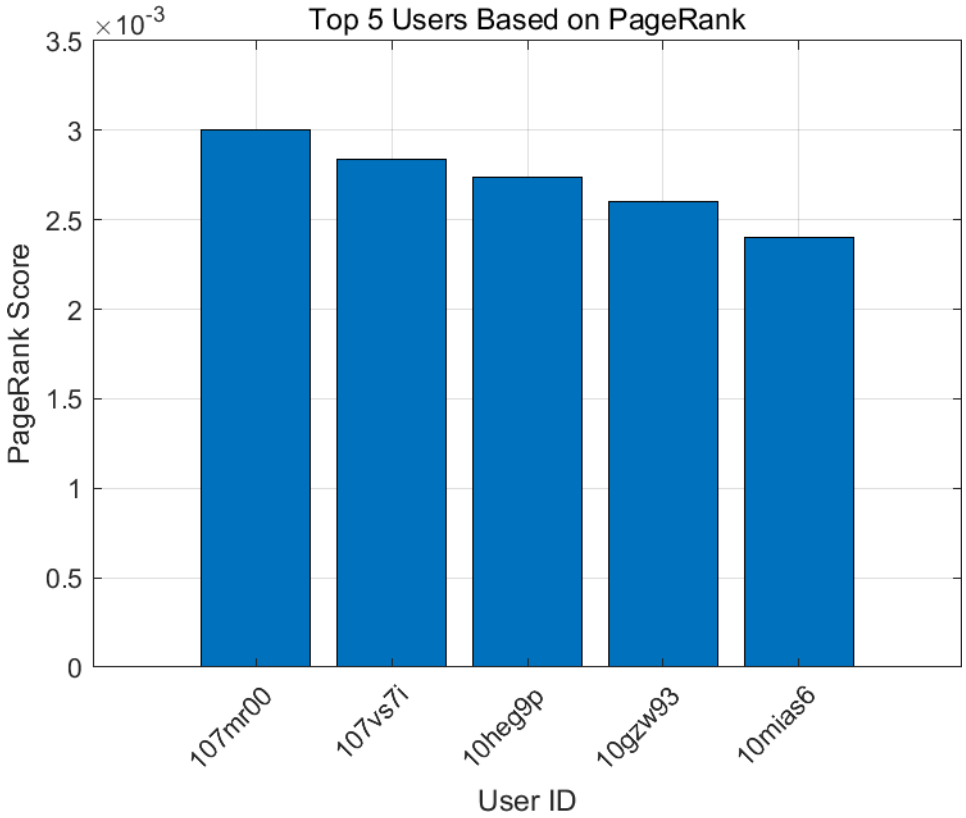

Figure 4.3 Top 5 Users Based on PageRank

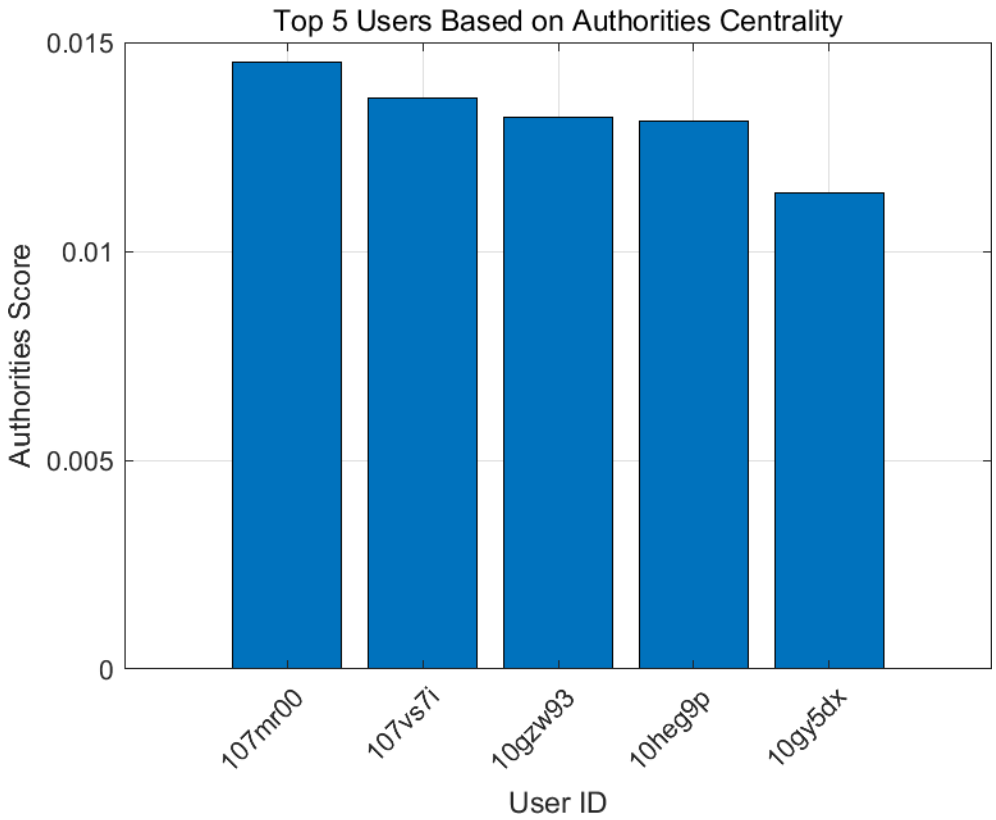

Figure 4.4 Top 5 Users Based on Authorities

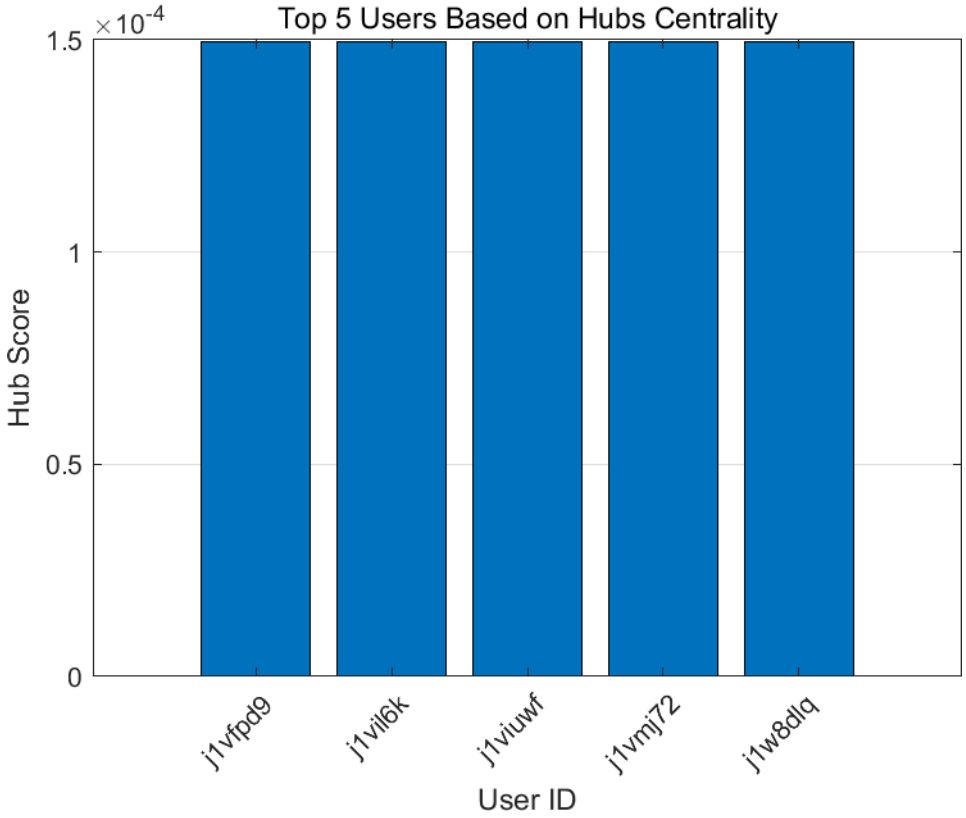

Figure 4.5 Top 5 Users Based on Hubs

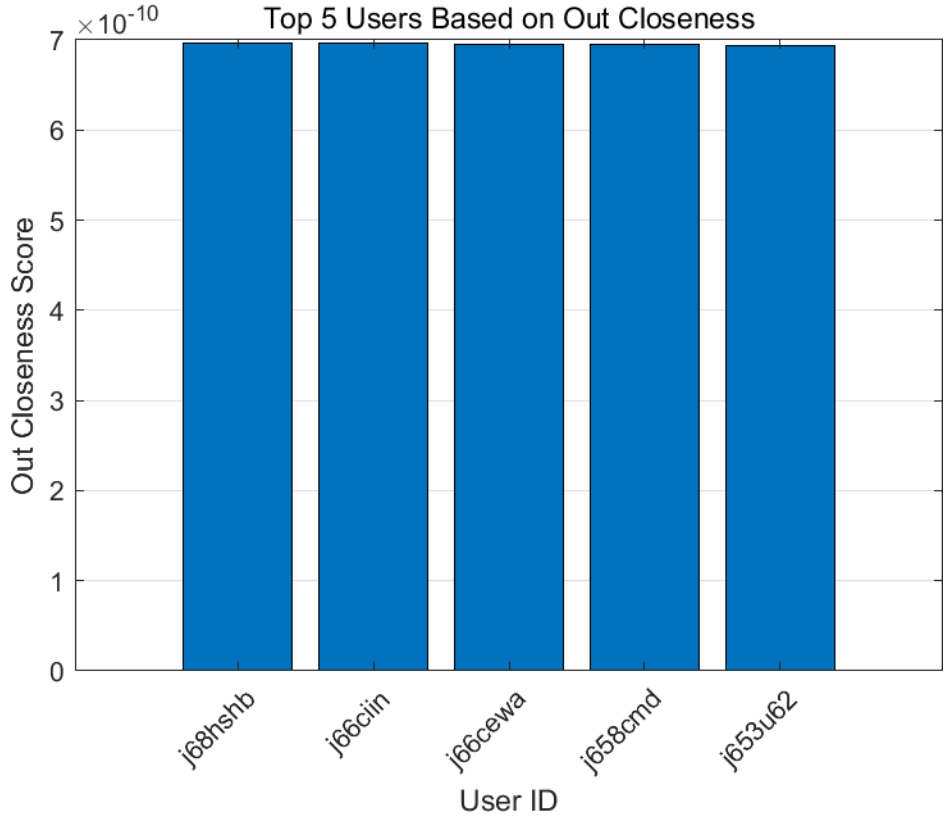

Figure 4.6 Top 5 Users Based on Out Closeness

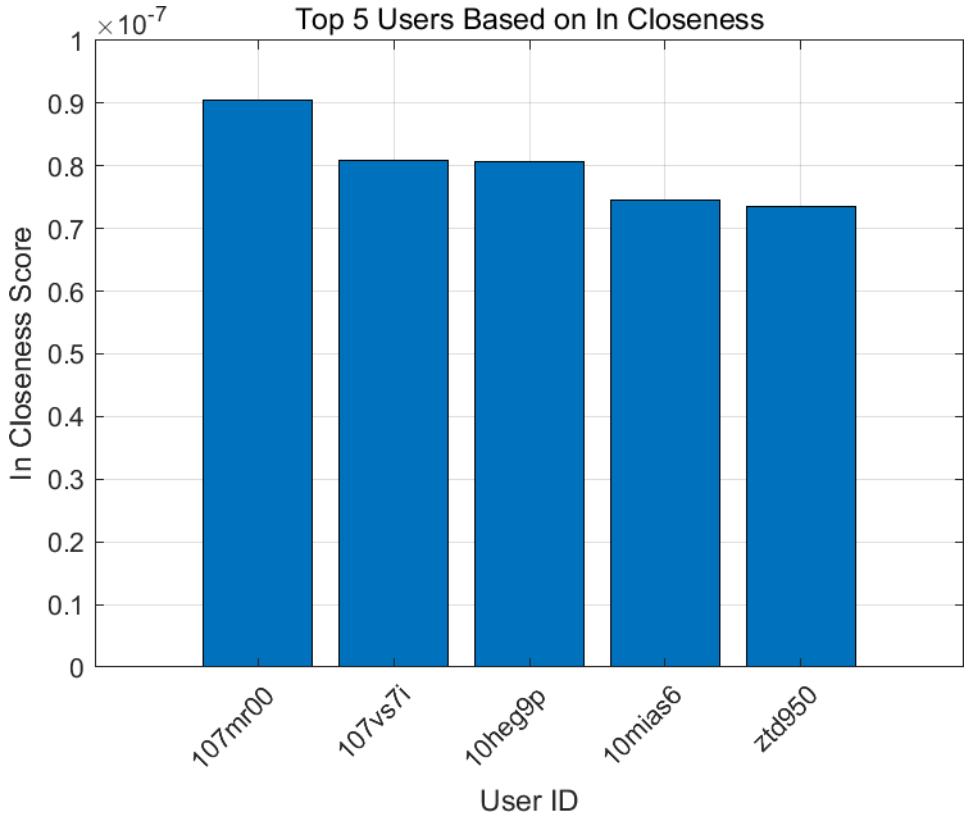

Figure 4.7 Top 5 Users Based on In Closeness

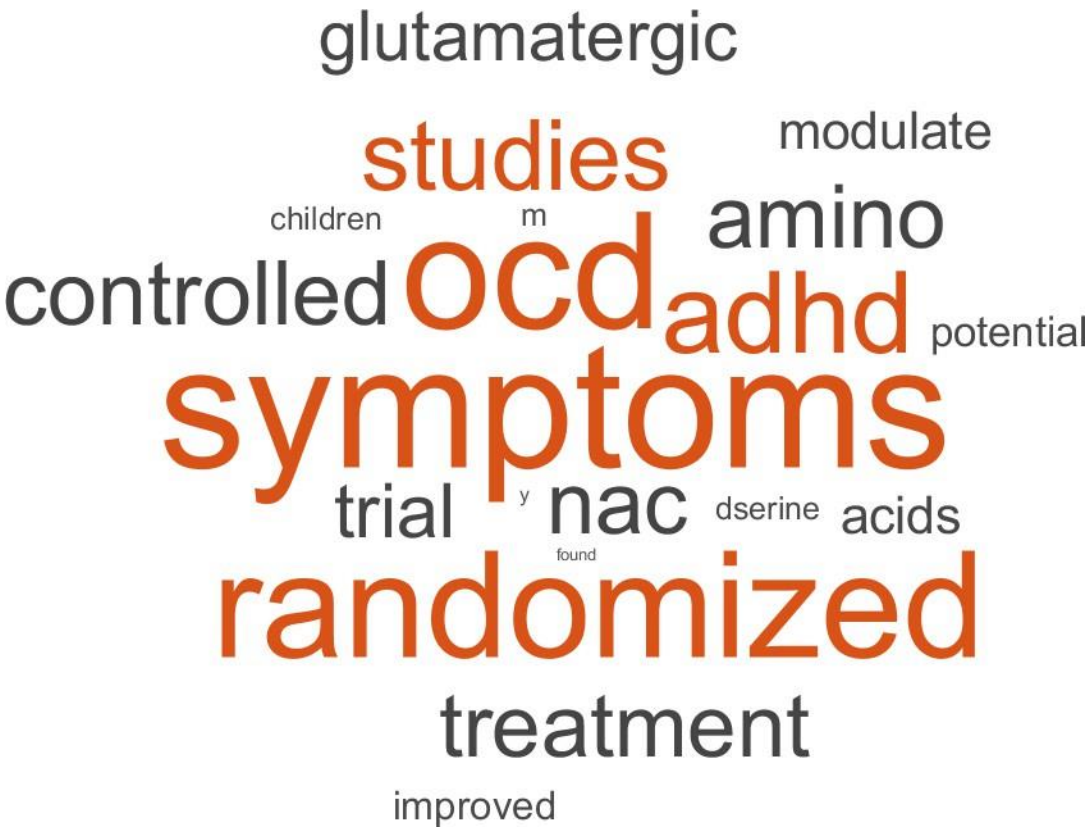

Figure 4.8 Word Cloud for Top 1 Users: 107mr00

Figure 4.9 Word Cloud for Top 2 Users: 107vs7

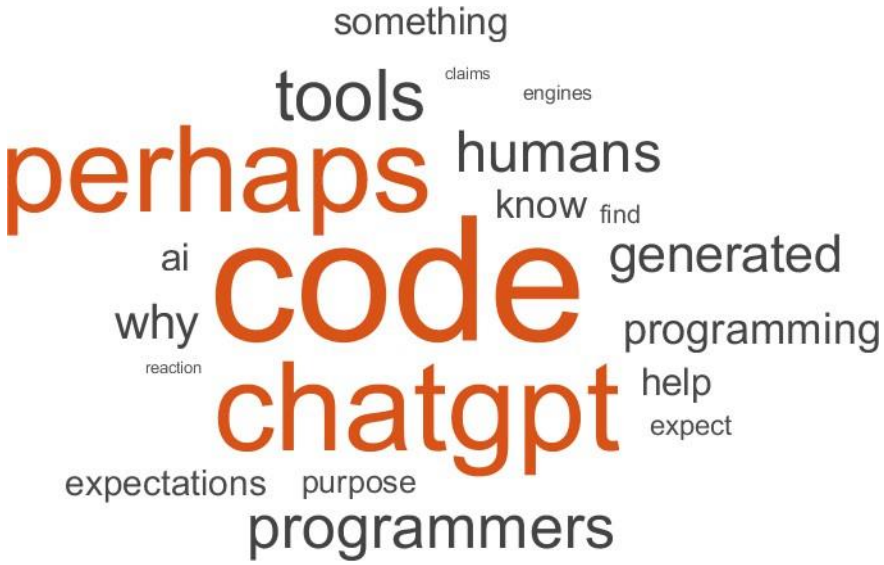

Figure 4.10 Word Cloud for Top 3 Users: 10heg9p

Figure 4.11 Word Cloud for Top 4 Users: 10gzw93

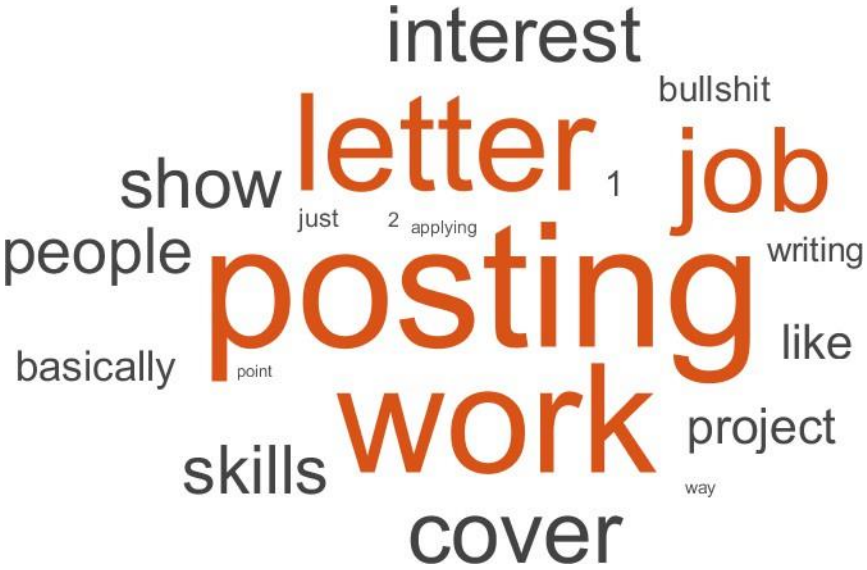

Figure 4.12 Word Cloud for Top 5 Users: 10mias6

Chapter 5. Conclusion

5.1 Summary

Key topics were evaluated and opinion leaders were identified based on a huge dataset of 52,416 comments. The research understands the primary subjects being addressed and the patterns of influence guiding these discussions were the two major goals of the research. This research examined conversations regarding ChatGPT on Reddit. The creation of a directed network graph was the method's main objective. This graph employed connections (parent-child relationships) and comments' unique IDs to show which comments replied to which comments. Each point in the network stands for a comment and the person who made it. The lines between the points show the order of the replies. This helped the researcher see how ideas and information moved in the conversation. To give a full picture, the researcher used two methods. One was impact analysis with PageRank. The other was topic analysis with LSA. This helped find the main topics and the users who led the talk. These results give useful information about what people talked about and how they took part in the ChatGPT group on Reddit.

5.2 Future Work

This study examined a single Reddit data snapshot and extracted important themes from all of the comments that were gathered. ChatGPT has made great progress in both development and popular awareness. Future studies can use dynamic topic modeling to see how discussion topics change over time. This method uses Dynamic Latent Dirichlet Allocation (DLDA). It helps track how topics grow. For example, it can show if people first talked about tech features, then later talked about ethics or real-life use. More details can be found by mixing topic modeling with sentiment analysis. This helps show if each topic has a positive, negative, or neutral tone.

REFERENCES

[1] Y. Lian, H.-B. Tang, M. Xiang, and X. Dong, “Public attitudes and sentiments toward ChatGPT in China: A text mining analysis based on social media,” *Technology in Society*, vol. 76, pp. 102442–102442, Dec. 2023.

[2] A. P. Mulia, P. R. Piri, and C. Tho, “Usability Analysis of Text Generation by ChatGPT OpenAI Using System Usability Scale Method,” *Procedia Computer Science*, vol. 227, pp. 381–388, Jan. 2023.

[3] L.-H. Hsieh, W.-C. Liao, and E.-Y. Liu, “Feasibility assessment of using ChatGPT for training case conceptualization skills in psychological counseling,” *Artificial Humans*, vol. 2, no. 2, p. 100083, Aug. 2024.

[4] M. Y. Hameed, L. Al-Hindi, S. Ali, H. K. Jensen, and C. C. Shoults, “Broadening the Understanding of Medical Students’ Discussion of Radiology Online: A Social Listening Study of Reddit,” *Current Problems in Diagnostic Radiology*, vol. 52, no. 5, pp. 377–382, Sep. 2023.

[5] A. M. Bunting, D. Frank, J. Arshonsky, M. A. Bragg, S. R. Friedman, and N. Krawczyk, “Socially-supportive norms and mutual aid of people who use opioids: An analysis of Reddit during the initial COVID-19 pandemic,” *Drug and Alcohol Dependence*, vol. 222, p. 108672, May 2021.

[6] C. Keerthigha, S. Singh, K. Q. Chan, and N. Caltabiano, “Helicopter parenting through the lens of Reddit: A text mining study,” *Heliyon*, vol. 9, no. 10, p. e20970, Oct. 2023.

[7] Y. Liu, S. Hu, H. Zhang, Q. Dong, and W. Liu, “Intelligent mining methodology of product field failure data by fusing deep learning and association rules for after-sales serEvnicgeineterxint,g” *Applications of Artificial Intelligence*, vol. 133, p. 108303, Jul. 2024.

[8] Z. Grotenhuis, P. J. Mosteiro, and A. M. Leeuwenberg, “Modest performance of text mining to extract health outcomes may be almost sufficient for high-quality prognostic model development,” *Computers in Biology and Medicine*, vol. 170, p. 108014, Mar. 2024.

[9] R. Nadlifatin *et al.*, “Exploring the Effectiveness of Work from Home: A Text Mining Analysis of Employee Perceptions and Experiences,” *Procedia Computer Science*, vol.

234, pp. 448–454, Jan. 2024.

[10] Elseline Senave, M. J. Jans, and R. P. Srivastava, “The application of text mining in accounting,” vol. 50, pp. 100624–100624, Sep. 2023.

[11] D. Xue and Z. Shao, “Patent text mining based hydrogen energy technology evolution path identification,” *International Journal of Hydrogen Energy*, vol. 49, pp. 699–710, Jan. 2024.

[12] M. Abdelsamie *et al.*, “Combining text mining, in situ characterization, and ab initio calculations to rationalize BiFeO3 crystallization pathways,” *Matter*, vol. 6, no. 12, pp. 4291–4305, Dec.2023.

[13] A. N. G. Puspita, I. Surjandari, and B. K. L. Kemala, “A Scientometric Analysis of electric vehicle-based text mining,” *Procedia Computer Science*, vol. 234, pp. 114–122, Jan. 2024.

[14] M. Elhaj *et al.*, “Analyzing pancreatic characteristics in diabetic patients: A texture-based CT investigation with volume assessment,” *Journal of Radiation Research and Applied Sciences*, vol. 17, no. 3, p. 100967, Sep. 2024.

[15] R. Iovanov *et al.*, “Testing the disintegration and texture-related palatability predictions for orodispersible tablets using an instrumental tool coupled with multivariate analysis: Focus on process variables and analysis settings,” *European Journal of Pharmaceutical Sciences*, vol. 198, p. 106801, Jul. 2024.

[16] J. Lin, Z. Xu, G. Sharma, and T. N. Pappas, “Texture representation via analysis and synthesis with generative adversarial networks,” *e-Prime - Advances in Electrical Engineering, Electronics and Energy*, vol. 6, p. 100286, Dec. 2023.

[17] P. Rowland, N. Cupido, M. Albert, and S. Kitto, "Morbidity and mortality rounds in the medical literature: A text based analysis of shifting knowledge regimes," *SSM - Qualitative Research in Health*, vol. 2, 2022, Art. no. 100169.

[18] S. Kim and W. S. Lee, "Network text analysis of medical tourism in newspapers using text mining: The South Korea case," *Tourism Management Perspectives*, vol. 31, pp. 332-339, 2019.

[19] K. Gao, P. Anandhan, and R. Kumar, "Analysis and evaluation of the regional air quality index forecasting based on web-text sentiment analysis method," *Environmental Impact Assessment Review*, vol. 87, 2021, Art. no. 106514.

[20] G. W. Alpers, A. J. Winzelberg, C. Classen, H. Roberts, P. Dev, C. Koopman, and C.

B. Taylor, "Evaluation of computerized text analysis in an Internet breast cancer support group," *Computers in Human Behavior*, vol. 21, no. 2, pp. 361-376, 2005.

[21] M. Sievert, O. Conrad, S. K. Mueller, R. Rupp, M. Balk, D. Richter, K. Mantsopoulos, H. Iro, and M. Koch, "Risk stratification of thyroid nodules: Assessing the suitability of ChatGPT for text- based analysis," *American Journal of Otolaryngology*, vol. 45, no. 2, 2024, Art. no. 104144.

[22] E. J. Bae, A. R. Jang, H. Park, and J. Y. Yoon, "Investigating knowledge structure and research trends in child and adolescent health literacy research through network text analysis," *Journal of Pediatric Nursing*, vol. 67, pp. 57-63, 2022.

[23] G. Guerrero, D. Avila, F. J. M. da Silva, A. Pereira, and A. Fernández-Caballero, "Internet- based identification of anxiety in university students using text and facial emotion analysis," *Internet Interventions*, vol. 34, 2023, Art. no. 100679.

[24] J. Lüdering and P. Tillmann, "Monetary policy on twitter and asset prices: Evidence from computational text analysis," *The North American Journal of Economics and Finance*, vol. 51, 2020, Art. no. 100875.

[25] R. Taborda, "Procedural transparency in Latin American central banks under inflation targeting schemes. A text analysis of the minutes of the Boards of Directors," *Ensayos sobre Política Económica*, vol. 33, no. 76, pp. 76-92, 2015.

[26] K. Zhu, L. Du, and Y. Feng, "Government attention on environmental protection and firms' carbon reduction actions: Evidence from text analysis of manufacturing enterprises," *Journal of Cleaner Production*, vol. 423, 2023, Art. no. 138703.

[27] R. Chen, P. Xu, H. Yao, and Y. Ding, "How will China achieve net-zero? A policy text analysis of Chinese decarbonization policies," *Energy Research & Social Science*, vol. 99, 2023, Art. no. 103051.

[28] J. Sun, Z. Shi, and Y. Zhou, "The impact of intellectual property articles on the quality of export products: Based on text analysis of China's trade agreements," *International Review of Economics & Finance*, vol. 92, pp. 1230-1249, 2024.

[29] C. Cheng, B. Huang, Y. Wang, and L. Hu, "Military background officials, risk awareness, and local government fiscal balance: Novel evidence from text analysis of Chinese local officials' news reports," *China Economic Review*, vol. 85, 2024, Art. no. 102166.

[30] S. Lin, Y. Kung, and F. Y. Leu, "Predictive intelligence in harmful news identification

by BERT-based ensemble learning model with text sentiment analysis," *Information Processing & Management*, vol. 59, no. 2, 2022, Art. no. 102872.

[31] M. Bogers, F. Biermann, A. Kalfagianni, and R. E. Kim, "Sustainable Development Goals fail to advance policy integration: A large-n text analysis of 159 international organizations," *Environmental Science & Policy*, vol. 138, pp. 134-145, 2022.

[32] Y. Zhang, W. Wang, L. Mi, Y. Liu, L. Qiao, G. Ni, and X. Wang, "Deconstructing the organizational resilience of construction firms in major emergencies: A text mining analysis of listed construction companies in China," *International Journal of Disaster Risk Reduction*, vol. 106, 2024, Art. no. 104473.

[33] I. Kong, F. O. Sarmiento, and L. Mu, "Crowdsourced text analysis to characterize the U.S. National Parks based on cultural ecosystem services," *Landscape and Urban Planning*, vol. 233, 2023, Art. no. 104692.

[34] W. Yu, B. Zhu, and X. Liu, "Text analysis of China's noise pollution prevention and control policy from the perspective of policy tools," *Environmental Technology & Innovation*, vol. 34, 2024, Art. no. 103569.

[35] X. Chen, Q. Xie, X. Cao, and Q. Li, "Examining the effectiveness of China's energy poverty alleviation policies: A text analysis on inter-provincial panel data," *Energy Policy*, vol. 186, 2024, Art. no. 113978.

[36] M. Vanhala, C. Lu, J. Peltonen, S. Sundqvist, J. Nummenmaa, and K. Järvelin, "The usage of large data sets in online consumer behaviour: A bibliometric and computational text-mining– driven analysis of previous research," *Journal of Business Research*, vol. 106, pp. 46-59, 2020.

[37] W. Liao, B. Zeng, J. Liu, P. Wei, X. Cheng, and W. Zhang, "Multi-level graph neural network for text sentiment analysis," *Computers & Electrical Engineering*, vol. 92, 2021, Art. no. 107096.

[38] F. Lin, S. Liu, C. Zhang, J. Fan, and Z. Wu, "StyleBERT: Text-audio sentiment analysis with Bi- directional Style Enhancement," *Information Systems*, vol. 114, 2023, Art. no. 102147.

[39] Y. Lin, P. Ji, X. Chen, and Z. He, "Lifelong Text-Audio Sentiment Analysis learning," *Neural Networks*, vol. 162, pp. 162-174, 2023.

[40] Y. Li, J. Chan, G. Peko, and D. Sundaram, "An explanation framework and method for AI- based text emotion analysis and visualisation," *Decision Support Systems*, vol.

178, 2024, Art. no. 114121.

[41] L. Celardo and M. G. Everett, "Network text analysis: A two-way classification approach," *International Journal of Information Management*, vol. 51, 2020, Art. no. 102009.

[42] H. Jang, S. Kim, and B. Yoon, "An eXplainable AI (XAI) model for text-based patent novelty analysis," *Expert Systems with Applications*, vol. 231, 2023, Art. no. 120839.

[43] H. T. Ha and A. Horák, "Information extraction from scanned invoice images using text analysis and layout features," *Signal Processing: Image Communication*, vol. 102, 2022, Art. no. 116601.

[44] Y. Kwon and J. Kwon, "An overview of Telecommunications Policy's 40-year research history: Text and bibliographic analyses," *Telecommunications Policy*, vol. 41, no. 10, pp. 878-890, 2017.

[45] J. R. H. Manning and L. Sarkisov, "Unveiling the synthesis patterns of nanomaterials: a text mining and meta-analysis approach with ZIF-8 as a case study," *Digital Discovery*, vol. 2, no. 6, pp. 1783-1796, 2023.

[46] V. Vasilyev, A. Vulfin, V. Gvozdev, and A. Nikonov, "Semantic Text Analysis Technology Application in Assessing Current Threats and Software Vulnerabilities," *IFAC-PapersOnLine*, vol. 54, no. 13, pp. 599-604, 2021.

[47] A. Razavi, "ChatGPT Reddit," *Kaggle*, 2022.

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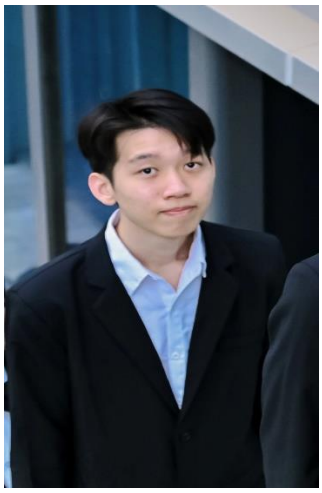

Cheng Lang is currently a senior student with a major in Computer Science and Technology in the College of Global Talents at Beijing Institute of Technology, Zhuhai (BITZH). During his undergraduate period, he completed many projects, such as the establishment of a student library resource service database and a Stadium management system web development. Lang proposed a study called "Identification of Topics and Opinion Leaders in Reddit Discussions on ChatGPT" when he was an undergraduate, which has high research value.

In the spring of 2025, Lang received postgraduate admission letters from University College London, the University of Manchester, and the University of Bristol. He will study for postgraduate studies at the University of Manchester.